b. Solve

(i) $9pqz^4 = 4\left(1+z^3\right)$ (6 Marks)

(ii) $\left(D^2+DD'-6D'^2\right)z = e^{3x+y}+x^2y$ (6 Marks)

29. a. Find the Fourier series of $f(x)=\frac{1}{2}(\pi-x)$ in $(0,2\pi)$ hence deduce that $1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\ldots.=\frac{\pi}{4}$.

(OR)

b. Determine the first two harmonics of the Fourier series for the following:

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	1.98	1.30	1.05	1.30	–0.88	–0.25	1.98

30. a. A tightly stretched string with fixed end points $x=0$, $x=l$ is initially at rest in its equilibrium position. If it is set vibrating giving each point a velocity $3x(l-x)$, find the displacement.

(OR)

b. A homogeneous rod of conducting material of length 100 cm has its ends kept at zero temperature and initially temperature is

$$u(x,0)=\begin{cases}x & ,\ 0\le x\le 50\\ 100-x, & 50\le x\le 100\end{cases}$$

Find the temperature $u(x,t)$ at any time t.

31. a. Find the Fourier transform of $f(x)$ given by

$$f(x)=\begin{cases}a^2-x^2, & if\ |x|<a\\ 0 & ,if\ |x|>a>0\end{cases}$$ hence prove that $\int_0^\infty \frac{\sin x-x\cos x}{x^3}dx=\frac{\pi}{4}$.

(OR)

b. Find the Fourier sine and cosine transform of x^{n-1} hence deduce that $\frac{1}{\sqrt{x}}$ is self reciprocal under both the transform.

32. a.i. Find $Z\left[\frac{1}{n(n-1)}\right]$. (6 Marks)

ii. Find the inverse Z-transform of $\frac{3z}{(z-1)(z-2)}$ using residue theorem. (6 Marks)

(OR)

b. Solve $y_{n+2}+4y_{n+1}+3y_n=3^n$ given $y_0=0$ and $y_1=1$ using Z-transform.

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Reg. No.														

B.Tech. DEGREE EXAMINATION, MAY 2023

Eighth Semester

15MA201 – TRANSFORMS AND BOUNDARY VALUE PROBLEMS

(For the candidates admitted during the academic year 2015-2016 to 2017-2018)

Note:

(i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.

(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer **ALL** Questions

1. The partial differential equation formed by eliminating arbitrary constants a and b from $z=ax^3+by^3$ is

(A) $z=px+qy$ (B) $3z=px+qy$

(C) $px+qy=0$ (D) $px=qy$

2. The singular integral of $z=px+qy+pq$ is

(A) $z+xy=0$ (B) $z-xy=0$

(C) $4z=y^2-x^2$ (D) $xy=1$

3. The particular integral of $\left(D^2+3DD'+4D'^2\right)z=e^{x-y}$ is

(A) $\frac{e^{x-y}}{8}$ (B) $\frac{e^{x-y}}{2}$

(C) $\frac{x}{2}e^{x-y}$ (D) $\frac{x^2}{2}e^{x-y}$

4. The general solution of $\left(D^2-2DD'+D'^2\right)z=0$ is

(A) $\phi_1(y+x)+\phi_2(y+x)$ (B) $\phi_1(y+x)+\phi_2(y+2x)$

(C) $\phi_1(y+x)+x\phi_2(y+x)$ (D) $\phi_1(y)+x\phi_2(y)$

5. The constant a_0 of the Fourier series for $f(x)=x$ in $0\le x\le 2\pi$ is

(A) π (B) 2π

(C) 3π (D) 0

6. The value of Fourier series of $f(x)$ in $0<x<2\pi$ at $x=0$ is

(A) $f(0)$ (B) $f(2\pi)$

(C) $\frac{f(0)+f(2\pi)}{2}$ (D) 0

7. The root mean square value of $f(x)=x^2$ in$(-\pi, \pi)$ is

(A) $\pi/\sqrt{5}$ (B) $\pi^2/\sqrt{5}$

(C) $\pi/25$ (D) $\pi^2/5$

8. If $f(x)=\sum_{n=1}^{\infty}b_n\sin\left(\frac{n\pi x}{l}\right)$ in $(0,l)$ then $\int_0^l (f(x))^2\,dx$ is

(A) $\frac{l}{2}\Sigma b_n^2$ (B) $\frac{l^2}{2}\Sigma b_n^2$

(C) $\frac{1}{4}\Sigma b_n^2$ (D) $\frac{l^2}{2}\Sigma b_n^2$

9. In one dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$, a^2 stands for

(A) T/m (B) m/T
(C) k/C (D) k/CP

10. How many initial and boundary conditions are required to solve $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$

(A) Two (B) Three
(C) Four (D) Five

11. The suitable solution of one dimensional heat equation is

(A) $u(x,t) = (Acoxpx + B\sin px)e^{-\alpha^2 p^2 t}$ (B) $u(x,t) = (Acoxpx + B\sin px)e^{\alpha^2 p^2 t}$
(C) $u(x,t) = \left(Ae^{px} + Be^{-px}\right)e^{-\alpha^2 p^2 t}$ (D) $u(x,t) = \left(Ae^{px} + Be^{-px}\right)e^{\alpha^2 p^2 t}$

12. The steady state temperature of a rod of length l whose ends $x = 0$ and $x = l$ are kept at 30°C and 40°C respectively is

(A) $u(x) = \frac{10x}{l} + 30$ (B) $u(x) = \frac{20x}{l} + 30$
(C) $u(x) = \frac{10x}{l} + 20$ (D) $u(x) = \frac{10x}{l}$

13. If $F\{f(x)\} = F(s)$ then $F\{f(x-a)\}$ is equal to

(A) $e^{ias}F(s)$ (B) $e^{iax}F(s)$
(C) $e^{-ias}F(s)$ (D) $e^{-iax}F(s)$

14. Fourier sine transform of e^{-ax} is

(A) $\sqrt{\frac{2}{\pi}}\left(\frac{a}{s^2+a^2}\right)$ (B) $\sqrt{\frac{2}{\pi}}\left(\frac{s}{s^2+a^2}\right)$
(C) $\sqrt{\frac{2}{\pi}}\left(\frac{s}{s^2-a^2}\right)$ (D) $\sqrt{\frac{2}{\pi}}\left(\frac{a}{s^2-a^2}\right)$

15. $F^{-1}[F(s)G(s)]$ is equal to

(A) $f(x)+g(x)$ (B) $f(x)-g(x)$
(C) $f(x)*g(x)$ (D) $f(x)\bullet g(x)$

16. Fourier transform of $e^{-x^2/2}$ is

(A) $e^{s^2/2}$ (B) $e^{i^2 s^2/4}$
(C) $e^{-s^2/2}$ (D) $e^{-i^2 s^2/4}$

17. Z-transform of $\sin\left(\frac{n\pi}{2}\right)$ is

(A) $\frac{z}{z^2-1}$ (B) $\frac{z^2}{z+1}$
(C) $\frac{z}{z^2+1}$ (D) $\frac{z^2}{z^2-1}$

18. Z-transform of n is

(A) $\frac{z}{(z+1)^2}$ (B) $\frac{z}{(z-1)^2}$
(C) $\frac{z}{z+1}$ (D) $\frac{z}{z-1}$

19. $z^{-1}\left(\frac{z}{(z-1)^2}\right)$ is

(A) n (B) $n+1$
(C) $n-1$ (D) $\frac{1}{n}$

20. Z-transform of $\frac{a^n}{n!}$

(A) e^{az} (B) $e^{a/z}$
(C) $e^{1/2}$ (D) $e^{-a/z}$

PART – B (5 × 4 = 20 Marks)
Answer **ANY FIVE** Questions

21. Form partial differential equation by eliminating arbitrary constants a, b, c from $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

22. Find the particular integral of $\left(D^2 - 3DD' + 2D'^2\right) = \sin(x-2y)$.

23. Obtain the half range sine series for $f(x) = x$ in $0 < x < 2$.

24. Write down the one dimensional wave equation and all its possible solution.

25. Prove that $F[f(ax)] = \frac{1}{a}F\left(\frac{s}{a}\right)$, $a > 0$.

26. Prove that $Z\left[\frac{1}{n+1}\right] = z\log\left(\frac{z}{z-1}\right)$

27. Using convolution theorem evaluate $Z^{-1}\left[\frac{z^2}{(z-1)(z-3)}\right]$.

PART – C (5 × 12 = 60 Marks)
Answer **ALL** Questions

28. a.i. Form the partial differential equation by eliminating the arbitrary function ϕ from the relation $\phi\left(x^2+y^2+z^2, xyz\right) = 0$. (6 Marks)

ii. Solve $(mz-ny)p + (nx-lz)q = ly - mx$. (6 Marks)

(OR)